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Master's Programme "Artificial Intelligence and Computer Vision"
Master of Science

REPORT

on the Final Project

Modernizing DeepSORT: Selectable Detectors, Re-Identification Models, and a Standalone Body-REID Identity System for Multi-Object Tracking

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Moscow, 2026

Contents

Abstract

This report describes the modernization of the DeepSORT multi-object tracker. The original DeepSORT consumes precomputed detections and a fixed appearance encoder; this work replaces those fixed components with selectable modern object detectors and re-identification (REID) models while keeping the Kalman-filter and Hungarian-assignment core unchanged. The detector and REID backends are wired behind two adapter interfaces with a lazy-import registry, so models from different source libraries can be selected before a run without touching the tracking core. A segmentation detector and a standalone body-REID identity component are added as optional modules.

The system is evaluated on six MOT-Challenge sequences (TUD-Campus, TUD-Stadtmitte, KITTI-17 and PETS09-S2L1 from the 2D MOT 2015 benchmark; MOT16-09 and MOT16-11 from MOT16) using the Higher-Order Tracking Accuracy (HOTA) metric computed with TrackEval. The unmodified baseline reaches a mean HOTA of 40.17. The best modern configuration (YOLOv8m detector with an OSNET REID model) reaches a mean HOTA of 52.54 (+12.37 absolute), exceeds the baseline on each of the six videos, and runs at 9.87 frames per second on a single Tesla T4 (above the 5 FPS real-time threshold). A per-video tuning study on TUD-Campus and an analysis of the environment-related build failures encountered during the project are reported, including the cases that did not work as expected.

Keywords: multi-object tracking, DeepSORT, person re-identification, HOTA, MOT-Challenge, YOLOv8, OSNet, segmentation.

1 General Description of the Project

1.1 Project Type

This is an applied engineering and research project. It involves re-engineering an existing open-source tracker into a configurable system, integrating several third-party detection and re-identification models, running controlled experiments on a public benchmark, and reporting both quantitative results and the failures encountered during development.

1.2 Context and Motivation

Multi-object tracking (MOT) assigns a consistent identity to each detected object across the frames of a video. DeepSORT [2] extends SORT [1] by adding an appearance descriptor to the motion-based association of the original tracker. The reference implementation, however, predates current detection and re-identification models: it reads precomputed detections from disk and uses a single fixed appearance encoder (`mars-small128`) [3]. The goal of this project is to modernize the tracker so that current detectors and re-identification models can be selected at run time, and to measure whether the modernized tracker improves tracking accuracy over the original baseline while remaining real-time.

1.3 Problem Statement

Given a sequence of frames, the tracker must output, for every frame, a set of bounding boxes each labelled with a persistent track identity. The integration contract of the original tracker is kept: each detection is represented as a `Detection(tlwh, confidence, feature)` object and passed to `tracker.predict()` followed by `tracker.update()`. The objective is to maximize the mean HOTA across the six evaluation videos, subject to a real-time constraint of at least 5 FPS for at least one configuration, while keeping the SORT core unchanged.

1.4 Dataset

All six evaluation sequences are taken from the training splits of the MOT-Challenge benchmarks, so ground-truth annotations are available for scoring. Four sequences come from 2D MOT 2015 [6] and two from MOT16 [7]. Table 1 summarizes them.

Table 1: Evaluation sequences. Frame counts are the standard MOT-Challenge values.

Sequence	**Benchmark**	**Frames**	**Scene**
TUD-Campus	2D MOT 2015	71	outdoor, crossing pedestrians
TUD-Stadtmitte	2D MOT 2015	179	outdoor, street
KITTI-17	2D MOT 2015	145	street, moving camera
PETS09-S2L1	2D MOT 2015	795	outdoor, surveillance
MOT16-09	MOT16	525	indoor, pedestrians
MOT16-11	MOT16	900	indoor, pedestrians

A 10-column versus 9-column difference in the ground-truth format of the two benchmarks is discussed in Section 6; it affected an early version of the data loader.

1.5 Evaluation Protocol and Metrics

The primary metric is HOTA [4], which combines a detection accuracy term (DetA) and an association accuracy term (AssA) into a single value and is computed with TrackEval [5]. MOTA, IDF1, DetA and AssA are reported as secondary metrics. Three screening procedures are used in addition to end-to-end HOTA:

- detector quality is scored independently as precision, recall and F1 against ground truth at an intersection-over-union threshold of 0.5;
- the re-identification model is screened by running the tracker on ground-truth boxes (the detector is replaced by ground truth), so that HOTA differences reflect appearance only;
- the standalone body-REID component is screened as a clustering problem on ground-truth crops using the Fowlkes-Mallows index, the silhouette coefficient and the Calinski-Harabasz index.

Speed is measured by `eval/fps_bench.py` with the detector, REID and tracker stages timed separately after a warm-up. Every run appends a row to `results/experiments.csv` so that the experimental history is reproducible.

2 Background and Related Work

2.1 SORT and DeepSORT

SORT [1] performs tracking-by-detection using a Kalman filter for motion prediction and the Hungarian algorithm for frame-to-frame assignment based on bounding-box overlap. DeepSORT [2] adds an appearance descriptor and a cosine distance gate, which reduces identity switches through occlusions. The appearance encoder of the reference implementation is the `mars-small128` cosine-metric-learning network [3].

2.2 Modern Detectors

Three detector families from three source repositories were integrated: YOLOv8 from Ultralytics [8], NanoDet [15], and RTMDet from MMDetection [13, 14]. A YOLOv8 segmentation variant is also integrated and converts the predicted mask to a bounding box.

2.3 Re-Identification Models

Re-identification models map a cropped person image to an embedding such that crops of the same person are close under cosine distance. OSNET [9] is a compact re-identification network; it is used here through the BoxMOT package [11], which bundles weights trained on MSMT17. A generic ImageNet backbone from `timm` [12] is used as a dependency-light fallback. The torchreid library [10] provides additional OSNET variants.

2.4 The HOTA Metric

HOTA [4] was proposed to balance detection and association in a single score, addressing known biases of MOTA (detection-dominated) and IDF1. It is the metric used by recent MOT-Challenge evaluations and is the primary target of this project.

3 Description of the Project Task Execution Process

3.1 Architecture and Integration Contract

The tracking core (`deep_sort/`: Kalman filter, track, tracker, linear assignment, IoU matching, detection) is kept unchanged except for one guard in the nearest-neighbour metric that makes the appearance distance dimension-agnostic, so that embeddings of different sizes (128, 512, $\sim$ 1280, 2048) can be used without further changes. Detection and re-identification are placed behind two abstract interfaces, `BaseDetector` and `BaseReIDExtractor`, each resolved by a lazy-import registry so that a backend that fails to install does not block the others. Figure 1 shows the data flow.

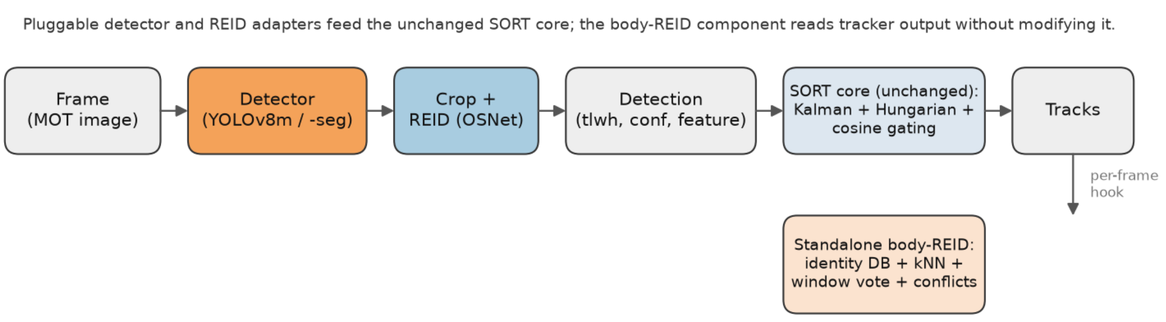

Figure 1: Data flow. Pluggable detector and REID adapters feed the unchanged SORT core; the standalone body-REID component reads tracker output through a per-frame hook without modifying the core.

3.2 Stage 0: Environment

The development environment is a Windows workstation for authoring and Google Colab Pro (Tesla T4, Python 3.12) for all compute. The notebook `notebooks/DeepSORT_Modern.ipynb` is a thin orchestrator: it clones the repository and calls the scripts in `eval/`, `data/` and `bodyreid/`, so that the logic lives in version-controlled modules rather than in notebook cells.

3.3 Stage 1: Data Preparation

`data/download_mot.py` downloads 2D MOT 2015 and MOT16 and lays them out in the directory structure expected by TrackEval. `data/prepare_gt_crops.py` exports ground-truth person crops labelled by track identity for the standalone body-REID study; it produced 21,105 crops over the six sequences.

3.4 Stage 2: Baseline Reproduction

`eval/run_baseline.py` reproduces the unmodified DeepSORT through the original `deep_sort_app.run` entry point using the provided detections and the `mars-small128` encoder with the original tracker settings. Its output is scored under the tracker name `baseline`.

3.5 Stage 3: Detector Study

`eval/det_eval.py` runs each detector over every sequence and reports precision, recall and F1 against ground truth at IoU ≥ 0.5.

3.6 Stage 4: Re-Identification Study

`eval/reid_eval.py` runs the tracker on ground-truth boxes and varies only the re-identification model, so the resulting HOTA reflects appearance association in isolation.

3.7 Stage 5: Full Pipeline and Per-Video Tuning

`eval/run_tracking.py` runs the complete live pipeline and writes MOT-format results that are scored by `eval/trackeval_runner.py`. Per-video parameters are set through sequence presets in `configs/sequences/`, which are merged automatically. The TUD-Campus preset was tuned with `eval/tune_tud_campus.py` (Section 4.5).

3.8 Stage 6: Standalone Body-REID and Segmentation

The standalone body-REID identity component (Section 5.2) and the YOLOv8 segmentation detector were added as optional modules and evaluated separately.

4 Description of the Project Results

All numbers in this section are from the live Colab runs recorded in `results/experiments.csv` and `report/results_table.md`. The headline tracking, re-identification and body-REID results are from the run of 2026-06-19; the detector and segmentation studies were refreshed on 2026-06-25 with the then-current YOLOv8m weights.

4.1 Baseline

Table 2 reports the per-video HOTA of the unmodified DeepSORT. Its mean of 40.17 is the threshold that every modern configuration must exceed.

Table 2: Per-video HOTA of the unmodified DeepSORT baseline.

	TUD-Campus	TUD-Stadt.	KITTI-17	PETS09	MOT16-09	MOT16-11	**Mean**
HOTA	39.86	36.75	43.41	44.84	36.24	39.95	**40.17**

4.2 Detector Evaluation

Figure 2 and Table 3 report detector precision, recall and F1. YOLOv8m has high mean recall (0.828) but its precision varies by sequence: it is high on the cleaner videos and lowest on the dense low-resolution KITTI-17 clip (precision 0.51), where it produces many false-positive boxes. NanoDet and RTMDet were not installed on the runtime (Section 6).

Table 3: Detector evaluation against ground truth (mean over six videos, IoU ≥ 0.5).

Detector	**Precision**	**Recall**	**F1**
YOLOv8m (box)	0.721	0.828	0.765
YOLOv8m-seg (mask-to-box)	0.727	0.827	0.768
NanoDet	not installed on this runtime		
RTMDet (MMDetection)	not installed on this runtime		

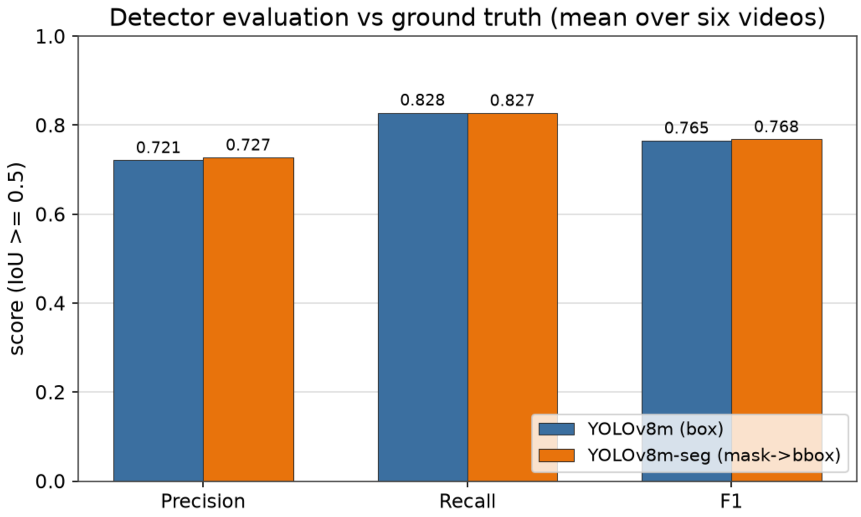

Figure 2: Detector precision, recall and F1 for the box and segmentation YOLOv8m variants.

4.3 Re-Identification Study

With detection fixed to ground truth, HOTA reflects association quality. OSNET is the strongest appearance model (mean HOTA 89.93), narrowly ahead of the `timm` backbone (89.22), followed by OSNET-AIN (87.33) and the smaller OSNET-x0.25 (86.13); see Figure 3. The high absolute values are expected because ground-truth boxes make detection near-perfect.

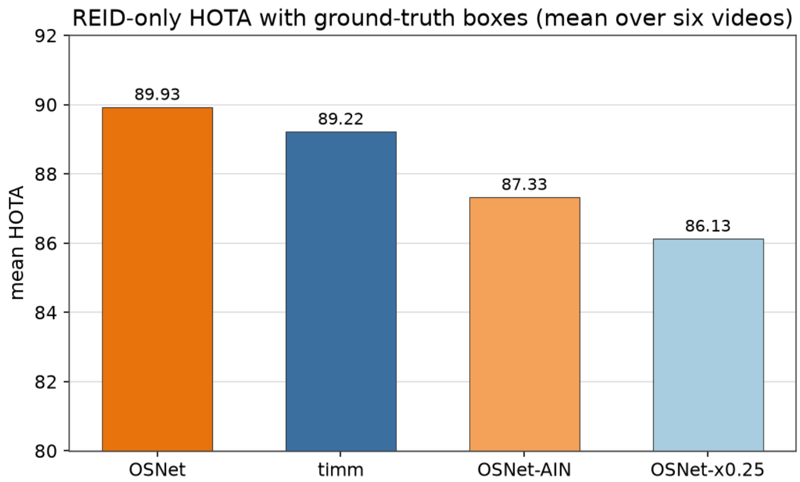

Figure 3: Re-identification-only HOTA with ground-truth boxes (mean over six videos).

4.4 Full Pipeline

Table 4 and Figure 4 report the live pipeline. The best configuration, YOLOv8m with OSNET, reaches a mean HOTA of 52.54 (+12.37 over the baseline) and exceeds the baseline on each of the six videos. The `timm` configuration reaches 47.70.

Table 4: Per-video HOTA: baseline and the two modern configurations. The YOLOv8m + OSNET row uses the tuned TUD-Campus preset (Section 4.5).

Configuration	TUD-Campus	TUD-Stadt.	KITTI-17	PETS09	MOT16-09	MOT16-11	**Mean**
Baseline	39.86	36.75	43.41	44.84	36.24	39.95	40.17
YOLOv8m + timm	37.57	57.63	45.82	50.89	45.56	48.73	47.70
YOLOv8m + OSNet	**46.98**	**59.62**	**48.75**	**60.28**	**48.57**	**51.07**	**52.54**

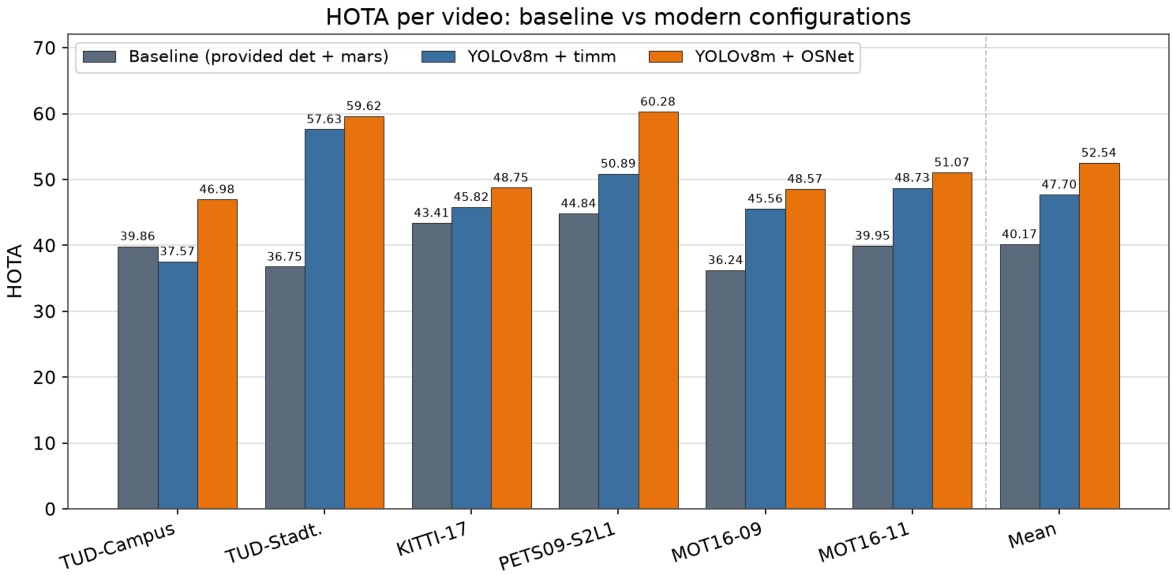

Figure 4: Per-video HOTA for the baseline and the two modern configurations.

4.5 TUD-Campus Tuning Case Study

TUD-Campus was the only video on which the modern configuration initially trailed the baseline (39.65 versus 39.86). Two sweeps were run with `eval/tune_tud_campus.py`, both expressed through the TUD-Campus sequence preset (the live runner has no command-line override mechanism).

- A recall-direction sweep (increasing `detector.imgsz` from 960 to 1280 and 1536, and lowering the confidence threshold) was not supported by the data: HOTA fell monotonically ($39.65 \rightarrow 35.0 \rightarrow 29.7$). Additional detections reduced the score.
- A precision-direction sweep was supported. The live pipeline applies no non-maximum suppression (`tracker.nms_max_overlap` default 1.0) and no separate confidence gate (`tracker.min_confidence` default 0.0), so the many low-confidence false positives on this dense clip are admitted to the tracker and create spurious tracks. Raising `detector.conf` to 0.45 reduces the false-positive count and raises HOTA to 46.98 (+7.1 over the baseline).

Table 5 and Figure 5 report the sweep. The change is confined to the TUD-Campus preset; the other five sequences are unchanged.

Table 5: TUD-Campus tuning sweep. Configurations above the baseline (39.86) are in the upper rows.

Configuration	HOTA	Configuration	HOTA
conf 0.45 (selected)	**46.98**	min_height 20	39.85
conf 0.35	44.61	control (conf 0.25)	39.65
nms 0.7 + conf 0.35	42.49	nms 0.7 alone	38.40
min_confidence 0.30	42.22	imgsz 1280	35.03
nms_max_overlap 0.5	41.96	imgsz 1536	29.75
nms 0.7 + minconf + h15	40.39		

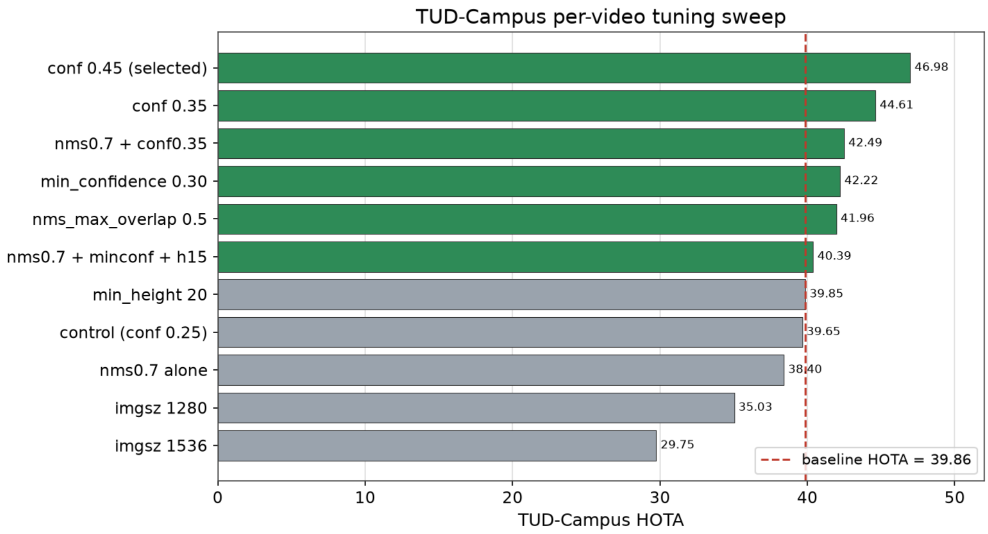

Figure 5: TUD-Campus tuning sweep. The dashed line is the baseline HOTA.

4.6 Metric Decomposition

Table 6 and Figure 6 decompose the improvement on the combined MOT16 split for YOLOv8m + timm. Most of the gain is on the detection term (DetA +12.30); the association term also improves (AssA +6.25). MOTA changes little, which is consistent with the fact that MOTA is dominated by detection recall.

Table 6: Metric decomposition on the combined MOT16 split (YOLOv8m + timm vs baseline).

Configuration	HOTA	MOTA	IDF1	DetA	AssA
Baseline (provided det + mars)	38.65	46.34	50.65	37.85	39.52
Modern (YOLOv8m + timm)	47.68	46.75	52.18	50.15	45.77
Difference	+9.03	+0.41	+1.53	+12.30	+6.25

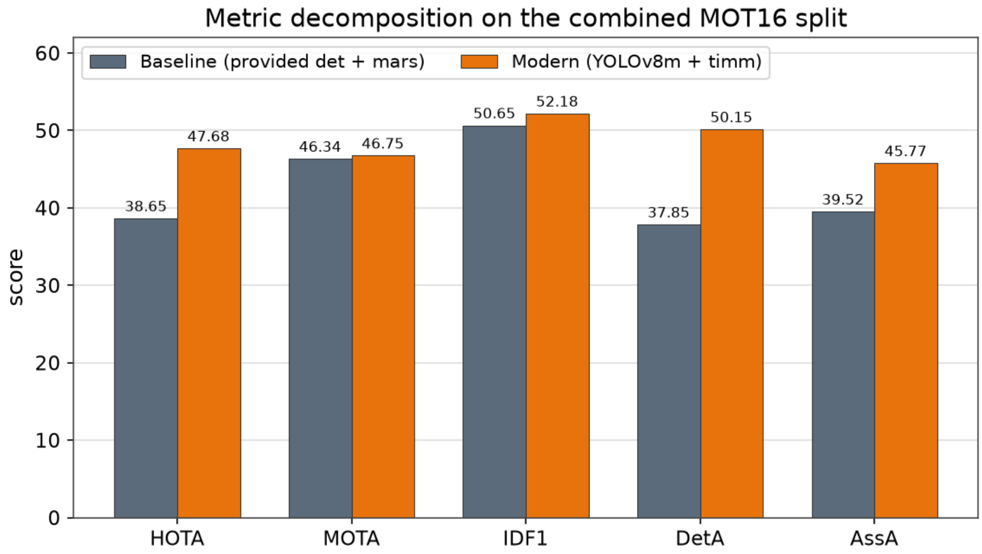

Figure 6: Metric decomposition on the combined MOT16 split.

4.7 Segmentation

The YOLOv8 segmentation detector produces a person mask and converts it to a tight bounding box. It is marginally better than the box detector on detection F1 (0.768 vs 0.765) and on mean HOTA (52.76 vs 52.54), but the mask head is substantially slower: on MOT16-09 the segmentation pipeline runs at 3.92 FPS, below the 5 FPS real-time threshold, against 9.87 FPS for the box pipeline. Segmentation stays real-time on the lighter 2D MOT 2015 clips ($\sim$10–14 FPS) but not on the heavier MOT16 sequences. The marginal accuracy gain therefore does not justify the loss of real-time throughput, so the box detector is the default and segmentation is a switchable option (Figure 7).

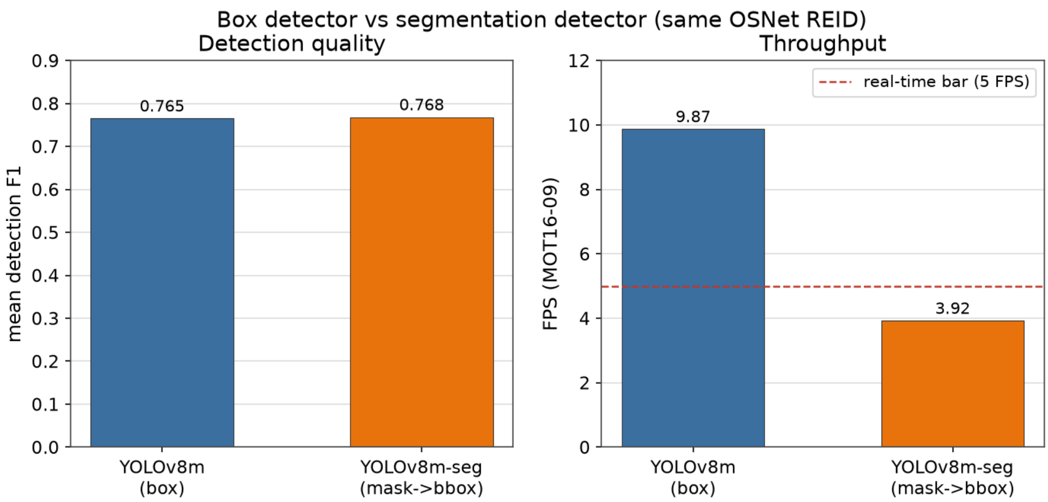

Figure 7: Box detector versus segmentation detector with the same OSNET re-identification model.

4.8 Standalone Body-REID

The standalone body-REID component was evaluated as a clustering problem on the 21,105 ground-truth crops (140 true identities) using OSNET descriptors. In the embedding mode (agglomerative clustering at the default threshold) the predicted number of identities is too high (420 vs 140, Fowlkes-Mallows index 0.630); in the assignment mode (a headless replay of the identity database) it is too low (30 vs 140, Fowlkes-Mallows index 0.242). The two modes bracket the true count, which indicates that the default thresholds should be tuned toward $n_{\text{pred}} \approx n_{\text{true}}$; the sweep utility `bodyreid/eval/sweep.py` exists for this purpose. Figure 8 shows the predicted counts against the true count.

Table 7: Standalone body-REID clustering metrics (OSNET descriptors, 21,105 crops, 140 true identities).

Mode	Fowlkes-Mallows	Silhouette	Calinski-Harabasz	$n_{\text{pred}}/n_{\text{true}}$
Embedding (agglomerative)	0.630	0.371	98.3	420 / 140
Assignment (DB replay)	0.242	–	–	30 / 140

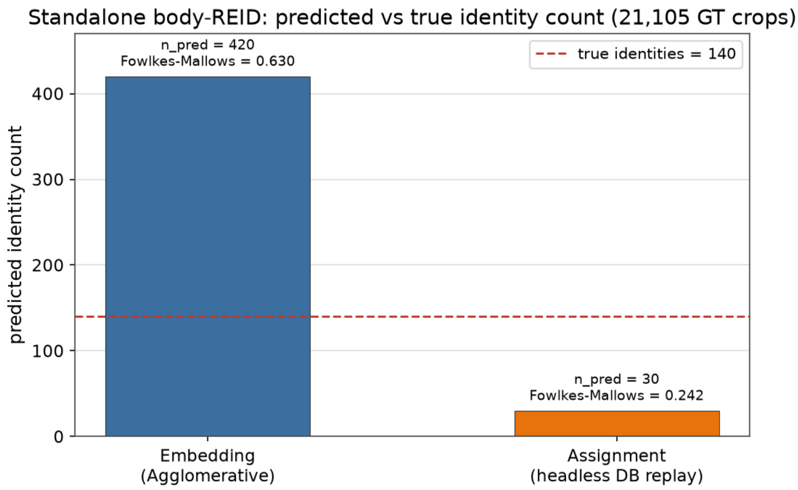

Figure 8: Standalone body-REID: predicted identity count in each mode against the true count.

4.9 Real-Time Performance

On MOT16-09, the YOLOv8m + OSNET configuration runs at 9.87 FPS overall, above the 5 FPS threshold. Figure 9 shows the per-frame time by stage: the detector and re-identification stages dominate, while the tracking core is comparatively cheap.

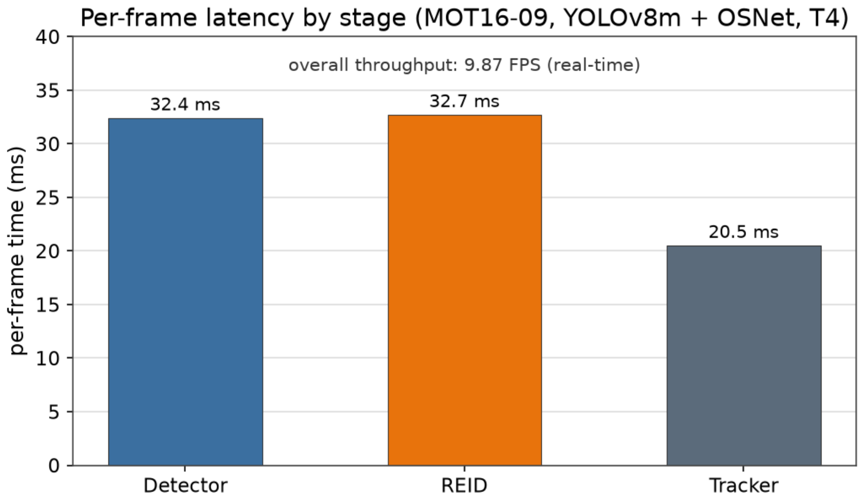

Figure 9: Per-frame latency by stage (MOT16-09, YOLOv8m + OSNET, Tesla T4).

5 Methods and Technologies

5.1 Detector and REID Adapters

Each detector adapter normalizes its output to person-only boxes in `tlwh` format with a confidence score; the COCO person class index, which differs between frameworks, is handled in the base class rather than hard-coded. Each REID adapter maps cropped images to L2-normalized embeddings; the cosine metric in the tracking core is dimension-agnostic, so different embedding sizes are interchangeable.

5.2 Standalone Body-REID Algorithm

The standalone identity component maintains a persistent identity database that is decoupled from the tracker's ephemeral track identities. Each identity keeps a running centroid and a capped gallery of recent descriptors. A new detection is matched by cosine distance with two thresholds: a match threshold below which the detection is assigned and enrolled, a higher threshold above which a new identity may be created only at track birth, and an intermediate band in which the detection is assigned but not enrolled. A per-track history is resolved over a time window by distance-weighted majority vote with hysteresis, and a cross-track conflict rule prevents two simultaneously active tracks from sharing one identity. The component is evaluated on its own terms as a clustering problem (Section 4); it reads tracker output through a per-frame hook and does not modify the tracking core.

5.3 Technology Stack

Table 8: Main software components.

Component	**Role**
PyTorch, Ultralytics	detector inference (YOLOv8m, YOLOv8-seg)
BoxMOT, torchreid, timm	re-identification backends
TrackEval	HOTA / MOTA / IDF1 scoring
scikit-learn	precision/recall/F1 and clustering metrics
NumPy, OpenCV	array and image operations
Google Colab Pro (Tesla T4)	compute environment

6 Deviations and Difficulties

This section records the problems encountered, including the cases that did not work. Most were environment issues on the bleeding-edge Colab runtime (Python 3.12, NumPy 2.x) rather than algorithm problems.

Table 9: Difficulties encountered and how they were handled.

Problem	Resolution
torchreid does not build under NumPy 2.x (metadata-generation failure).	The torchreid path to OSNET was blocked; OSNET was obtained through BoxMOT instead.
BoxMOT initially failed to build and is an optional dependency.	It is installed in a guarded step (`pip install boxmot || echo ...`) and excluded from the hard requirements set, so its failure cannot abort the core install. It built successfully in the final run, which unblocked OSNET.
A dependency-light REID was needed as a fallback.	A `timm` ImageNet backbone (not re-identification-trained) is used; it installs without a native build and still exceeds the baseline.
TrackEval uses removed NumPy aliases (`np.float`, `np.int`, `np.bool`).	The TrackEval source is patched in place at run time before scoring.
2D MOT 2015 ground truth is 10-column (columns 7–9 are world coordinates), while MOT16 is 9-column (class and visibility).	The loader reads class and visibility only when there are exactly nine columns; a regression test fixes both formats. The earlier code silently dropped the four 2D MOT 2015 sequences.
The official MOT-Challenge host was unreachable from Colab.	A mirror that preserves the MOT-Challenge directory layout is used; the official URLs remain configurable.
The notebook used IPython shell-variable substitution across cells, which expanded to empty values.	A configuration cell exports the values to the shell environment, so the `!` cells receive real environment variables.

7 Conclusion

The modernized DeepSORT replaces the fixed detector and appearance encoder of the reference implementation with selectable modern models behind two adapter interfaces, while keeping the SORT core unchanged. On the six MOT-Challenge evaluation videos, the best configuration (YOLOv8m with an OSNET re-identification model) reaches a mean HOTA of 52.54 versus 40.17 for the unmodified baseline (+12.37), exceeds the baseline on every video, and runs at 9.87 FPS on a Tesla T4. The improvement is driven mainly by detection quality (DetA +12.30) and, to a smaller degree, by association (AssA +6.25). A per-video tuning study on TUD-Campus showed that the binding constraint there was an excess of low-confidence detections rather than insufficient recall, and that raising the detector confidence threshold closed the gap. The standalone body-REID component is implemented and evaluated as a clustering problem; its default thresholds bracket the true identity count and are a clear target for the existing parameter sweep.

7.1 Future Directions

The threshold sweep for the standalone body-REID component, the measurement of its effect on the end-to-end tracking metrics, and the installation of the NanoDet and RTMDet detectors on a compatible runtime are the natural next steps. A torchreid build compatible with NumPy 2.x would allow the remaining OSNET variants and a ResNet-50 re-identification model to be measured in the same study.

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